Internet & network security

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Data communication & networking
FIREWALL

NETWORK SECURITY TERMINOLOGIES



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MALWARE

It is a type of malicious program or software which can damage your system



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Application Layer Protocols

Protocol	Port	Use
13. HTTP (HyperText Transfer Protocol)	80	Web page access.
14. HTTPS (HTTP Secure)	443	Secure web page access using SSL/TLS.
15. FTP (File Transfer Protocol)	21	File transfer between client and server.
16. SFTP (SSH File Transfer Protocol)	22	Secure file transfer over SSH.
17. TFTP (Trivial File Transfer Protocol)	69	Simple, unsecured file transfer.
18. SMTP (Simple Mail Transfer Protocol)	25	Sending emails.
19. POP3 (Post Office Protocol 3)	110	Receiving emails.
20. IMAP (Internet Message Access Protocol)	143	Email retrieval and management.
21. DNS (Domain Name System)	53	Resolving domain names to IP addresses.
22. DHCP (Dynamic Host Configuration Protocol)	67/68	Automatic IP assignment.

mail Rsv.

Application Layer Protocols

22. DHCP (Dynamic Host Configuration Protocol)	67/68	Automatic IP assignment.
23. SNMP (Simple Network Management Protocol)	161/162	Network device monitoring and management.
24. LDAP (Lightweight Directory Access Protocol)	389	Accessing directory services.
25. NTP (Network Time Protocol)	123	Time synchronization.
26. Telnet	23	Remote text-based access (insecure).
27. SSH (Secure Shell)	22	Secure remote access and command execution.
28. IRC (Internet Relay Chat)	194	Real-time chat communication.
29. SIP (Session Initiation Protocol)	5060	VoIP signaling.
30. RTP (Real-time Transport Protocol)	Dynamic	Media streaming.

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IoT

Application Layer Protocols

32. MQTT (Message Queuing Telemetry Transport) 1883 Lightweight messaging for IoT.
33. CoAP (Constrained Application Protocol) 5683 IoT communication in constrained devices.
34. AMQP (Advanced Message Queuing Protocol) 5672 Messaging middleware protocol.
35. SMB (Server Message Block) 445 File sharing in Windows networks.
36. NFS (Network File System) 2049 File sharing in Unix/Linux networks.
37. POP3S 995 Secure POP3 over SSL.
38. IMAPS 993 Secure IMAP over SSL.
39. LDAPS 636 Secure LDAP over SSL.
40. RDP (Remote Desktop Protocol) 3389 Remote desktop access for Windows.

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Application Layer Protocols

41. VNC (Virtual Network Computing) 5900 Remote desktop access.
42. SIP-TLS 5061 Secure VoIP signaling.
43. H.323 Dynamic VoIP and video conferencing.
44. XMPP (Extensible Messaging and Presence Protocol) 5222 Instant messaging.
45. Syslog 514 Logging messages from devices.
46. NetBIOS 137-139 Legacy name resolution and file sharing.
47. PPTP (Point-to-Point Tunneling Protocol) 1723 VPN tunneling.
48. L2TP (Layer 2 Tunneling Protocol) 1701 VPN tunneling.
49. IPSec N/A Secure network communications (VPN).
50. SCTP 989/990 Secure data transport in telecom networks.

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Protocol	Port / Layer	Use
1. IP (Internet Protocol)	N/A	Routing packets across networks.
2. ICMP (Internet Control Message Protocol)	N/A	Sending error and control messages (ping, traceroute).
3. ARP (Address Resolution Protocol)	N/A	Resolving MAC addresses from IP addresses.
4. RARP (Reverse ARP)	N/A	Resolving IP addresses from MAC addresses.
5. OSPF (Open Shortest Path First)	N/A	Dynamic routing in IP networks.
6. RIP (Routing Information Protocol)	N/A	Simple dynamic routing protocol.
7. BGP (Border Gateway Protocol)	N/A	Routing between autonomous systems on the internet.

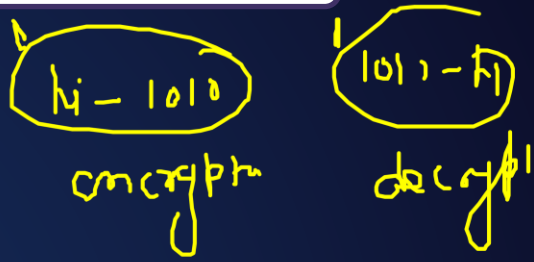


9. IGMP (Internet Group Management Protocol)	N/A	Managing multicast group memberships.
ICMPv6	N/A	Error reporting and diagnostics in IPv6 networks.

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Protocol	Port	Use
10. TCP (Transmission Control Protocol)	N/A	Reliable, connection-oriented data delivery.
11. UDP (User Datagram Protocol)	N/A	Fast, connectionless data delivery.
12. SCTP (Stream Control Transmission Protocol)	N/A	Reliable message-oriented transport, used in telecoms.

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Types of Encryption Techniques

A. Symmetric Encryption (Private-Key Encryption)

- **Concept:** Same key is used for both encryption and decryption.
- **Advantages:** Fast and efficient for large data.
- **Disadvantages:** Key distribution problem (both parties must securely share the key).
- **Common Algorithms:**
  - **DES (Data Encryption Standard)** – older standard, uses 56-bit key.
  - **AES (Advanced Encryption Standard)** – modern standard, supports 128, 192, 256-bit keys.
  - **RC4, Blowfish** – other symmetric algorithms.

Asymmetric Encryption (Public-Key Encryption)
Concept: Uses **two keys**:
Public Key: Shared openly, used to encrypt data.
Private Key: Kept secret, used to decrypt data.
Advantages: Solves the key distribution problem; more secure.
Disadvantages: Slower than symmetric encryption; computationally intensive.
Common Algorithms:
RSA (Rivest–Shamir–Adleman)
ECC (Elliptic Curve Cryptography)
DSA (Digital Signature Algorithm)

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Hash Functions (One-Way Encryption)
Converts data into a **fixed-length hash** that **cannot be reversed**.
Used for **data integrity and passwords**.
Common Algorithms: MD5, SHA-1, SHA-256

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. What is the main purpose of encryption?
a) To compress data
b) To make data unreadable to unauthorized users
c) To increase data size
d) To transfer data faster

Decryption is the process of:

a) Converting ciphertext into plaintext

b) Converting plaintext into ciphertext

c) Compressing the data

d) Sending data over the network

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Which of the following is **not a component** of an encryption system?

a) Key
b) Ciphertext
c) Firewall
d) Algorithm

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Symmetric vs Asymmetric Encryption

Q4. In symmetric encryption:

a) Different keys are used for encryption and decryption

b) The same key is used for both encryption and decryption

c) Only public key is used

d) No key is required

Asymmetric encryption uses:
a) Only one key
b) Two keys (public and private)
c) No key
d) Same key for all users

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